2.4 Algebraic Fractions

An algebraic fraction is a fraction that contains variables. For example,

$$\frac{x}{3}, \qquad \frac{2a}{5b}, \qquad \frac{x+1}{x-2}$$

are all algebraic fractions. Although algebraic fractions contain variables, they follow exactly the same rules as ordinary numerical fractions. For this reason, it is useful to think of algebraic fractions simply as ordinary fractions that happen to contain letters.

Definition

An algebraic fraction is a fraction whose numerator, denominator, or both contain algebraic expressions.

2.4.1 Simplifying Algebraic Fractions

Algebraic fractions can often be simplified by cancelling common factors. Remember that only factors can be cancelled.

Worked Example

Simplify

$$\frac{6x}{3}$$

Divide numerator and denominator by :

Worked Example

Simplify

$$\frac{12x^2}{4x}$$

Simplify the numerical factors:

Simplify the variables:

Therefore,

2.4.2 Factorisation Before Simplification

Many algebraic fractions cannot be simplified immediately. In these cases, the numerator or denominator must first be factorised.

Worked Example

Simplify

$$\frac{x^2 + 2x}{x}$$

Factorise the numerator:

Thus,

$$\frac{x^2 + 2x}{x} =$$

Worked Example

Simplify

$$\frac{3x + 12}{3}$$

Factorise the numerator:

Cancel the common factor :

Definition

Always factorise before attempting to cancel. Cancelling terms is not allowed. Only common factors may be cancelled.

2.4.3 Multiplying Algebraic Fractions

Algebraic fractions are multiplied in exactly the same way as ordinary fractions.

Definition

To multiply fractions:

1. Multiply the numerators.
2. Multiply the denominators.
3. Simplify if possible.

Worked Example

Simplify

$$\frac{x}{3} \times \frac{2}{5}$$

Multiply the numerators:

Multiply the denominators:

Therefore,

$$\frac{x}{3} \times \frac{2}{5} =$$

Worked Example

Simplify

$$\frac{3x}{4} \times \frac{8}{9}$$

Cancel common factors first:

Further simplification gives

2.4.4 Dividing Algebraic Fractions

Division of fractions uses reciprocals.

Definition

To divide fractions:

1. Keep the first fraction.
2. Change division to multiplication.
3. Flip the second fraction.

Worked Example

Simplify

$$\frac{x}{4} \div \frac{3}{5}$$

Rewrite as multiplication:

Multiply:

Worked Example

Simplify

$$\frac{2x}{3} \div \frac{4}{9}$$

Rewrite as a multiplication:

2.4.5 Adding Algebraic Fractions

Fractions can only be added directly if they have the same denominator.

Worked Example

Simplify

$$\frac{x}{7} + \frac{3}{7}$$

The denominators are already equal. Therefore,

If the denominators are different, a common denominator is required.

Worked Example

Simplify

$$\frac{x}{2} + \frac{1}{3}$$

The lowest common denominator is . Rewrite each fraction:

$$\frac{x}{2} + \frac{1}{3} =$$

2.4.6 Subtracting Algebraic Fractions

Subtraction follows the same idea.

Worked Example

Simplify

$$\frac{x}{4} - \frac{1}{4}$$

The denominators are already equal. Therefore,

Worked Example

Simplify

$$\frac{7}{x+1} - \frac{x}{3}$$

Write both fractions with denominator :

$$\frac{7}{x+1} - \frac{x}{3} =$$

2.4.7 Common Mistakes

Common errors involving algebraic fractions include:

- Cancelling terms instead of factors.
- Forgetting to factorise first.
- Multiplying numerators but not denominators.
- Forgetting to use the reciprocal when dividing.
- Adding denominators directly.
- Losing brackets.

Worked Example

A common mistake is

$$\frac{x+2}{x} = 2$$

This is incorrect. The x is not a factor of the entire numerator. The correct method is

$$\frac{x+2}{x} = \frac{x}{x} + \frac{2}{x} = 1 + \frac{2}{x}$$

2.4.8 Engineering Application

Engineering Application

Algebraic fractions occur frequently in engineering.

For example, the lens formula is

$$\frac{1}{f} = \frac{1}{u} + \frac{1}{v}$$

where f is the focal length, u is the object distance and v is the image distance.

Electrical engineering provides another example:

$$\frac{1}{R} = \frac{1}{R_1} + \frac{1}{R_2}$$

which describes resistors connected in parallel.

The ability to manipulate algebraic fractions is therefore essential when rearranging and using engineering formulae.

2.4.9 Exercises

Exercise

1. $$\frac{x+11}{2x-5}=2$$

2. $$\frac{5x-24}{x-4}=3$$

3. $$4y-9=\frac{2y+3}{3}$$

4. $$\frac{x+4}{3}+\frac{x+1}{2}=1$$

5. $$\frac{2x-5}{7}-\frac{2x-1}{2}=3$$

6. $$\frac{x+1}{2}+\frac{2x-1}{4}+\frac{x+2}{3}=1$$

7. $$\frac{z-7}{3}+\frac{2z-1}{4}=\frac{z+3}{6}$$

8. $$\frac{3x+2}{5}=4$$

9. $$\frac{x-3}{4}+\frac{x+1}{4}=2$$

10. $$\frac{2x+5}{3}=\frac{x-1}{2}$$

11. $$\frac{3y+4}{2}-\frac{y-2}{3}=5$$

12. $$\frac{x+2}{x}=3$$

13. $$\frac{2x-1}{x+3}=1$$

14. $$\frac{a+5}{4}+\frac{a-1}{6}=2$$

Answers to Exercise 2.4.9

1.

$$\begin{aligned}
\frac{x+11}{2x-5} &= 2 && \text{[multiply both sides by } 2x-5\text{]}\\
x+11 &= 2(2x-5) && \text{[multiply out the brackets]}\\
x+11 &= 4x-10 && \text{[subtract } 4x \text{ from both sides]}\\
-3x+11 &= -10 && \text{[subtract 11 from both sides]}\\
-3x &= -21 && \text{[divide both sides by -3]}\\
\therefore \qquad x &= 7
\end{aligned}$$

2.

$$\begin{aligned}
\frac{5x-24}{x-4} &= 3 && \text{[multiply both sides by } x-4\text{]}\\
5x-24 &= 3(x-4) && \text{[multiply out the brackets]}\\
5x-24 &= 3x-12 && \text{[subtract } 3x \text{ from both sides]}\\
2x-24 &= -12 && \text{[add 24 to both sides]}\\
2x &= 12 && \text{[divide both sides by 2]}\\
\therefore \qquad x &= 6
\end{aligned}$$

3.

$$\begin{aligned}
4y-9 &= \frac{2y+3}{3} && \text{[multiply both sides by 3]}\\
3(4y-9) &= 2y+3 && \text{[multiply out the brackets]}\\
12y-27 &= 2y+3 && \text{[subtract } 2y \text{ from both sides]}\\
10y-27 &= 3 && \text{[add 27 to both sides]}\\
10y &= 30 && \text{[divide both sides by 10]}\\
\therefore \qquad y &= 3
\end{aligned}$$

4.

$$
\begin{aligned}
\frac{x+4}{3}+\frac{x+1}{2} &= 1 && \text{[multiply both sides by 3]}\\
x+4+\frac{3(x+1)}{2} &= 3 && \text{[multiply both sides by 2]}\\
2(x+4)+3(x+1) &= 6 && \text{[multiply out the brackets]}\\
2x+8+3x+3 &= 6 && \text{[collect like terms]}\\
5x+11 &= 6 && \text{[subtract 11 from both sides]}\\
5x &= -5 && \text{[divide both sides by 5]}\\
\therefore \qquad x &= -1
\end{aligned}
$$

5.

$$
\begin{aligned}
\frac{2x-5}{7}-\frac{2x-1}{2} &= 3 && \text{[multiply both sides by 7]}\\
2x-5-\frac{7(2x-1)}{2} &= 21 && \text{[multiply both sides by 2]}\\
2(2x-5)-7(2x-1) &= 42 && \text{[multiply out the brackets]}\\
4x-10-14x+7 &= 42 && \text{[collect like terms]}\\
-10x-3 &= 42 && \text{[add 3 to both sides]}\\
-10x &= 45 && \text{[divide both sides by -10]}\\
\therefore \qquad x &= -4.5
\end{aligned}
$$

6.

$$\begin{aligned}
\frac{x+1}{2}+\frac{2x-1}{4}+\frac{x+2}{3} &= 1 && \text{[multiply both sides by 2]}\\
x+1+\frac{2(2x-1)}{4}+\frac{2(x+2)}{3} &= 2 && \text{[multiply both sides by 4]}\\
4(x+1)+2(2x-1)+\frac{8(x+2)}{3} &= 8 && \text{[multiply both sides by 3]}\\
12(x+1)+6(2x-1)+8(x+2) &= 24 && \text{[multiply out the brackets]}\\
12x+12+12x-6+8x+16 &= 24 && \text{[collect like terms]}\\
32x+22 &= 24 && \text{[subtract 22 from both sides]}\\
32x &= 2 && \text{[divide both sides by 32]}\\
\therefore \qquad x &= \frac{2}{32}=\frac{1}{16}
\end{aligned}$$

7. We could follow the same procedure as above, or multiply through by the lowest common denominator:

$$\begin{aligned}
\frac{z-7}{3}+\frac{2z-1}{4} &= \frac{z+3}{6} && \text{[multiply both sides by 12]}\\
12\left(\frac{z-7}{3}+\frac{2z-1}{4}\right) &= 12\left(\frac{z+3}{6}\right) && \text{[multiply out the brackets]}\\
4(z-7)+3(2z-1) &= 2(z+3) && \text{[multiply out the brackets]}\\
4z-28+6z-3 &= 2z+6 && \text{[collect like terms]}\\
10z-31 &= 2z+6 && \text{[subtract } 2z \text{ from both sides]}\\
8z-31 &= 6 && \text{[add 31 to both sides]}\\
8z &= 37 && \text{[divide both sides by 8]}\\
\therefore \qquad z &= \frac{37}{8}
\end{aligned}$$

8.

$$\begin{aligned}
\frac{3x+2}{5} &= 4 && \text{[multiply both sides by 5]}\\
3x+2 &= 20 && \text{[subtract 2]}\\
3x &= 18 && \text{[divide by 3]}\\
\therefore \quad x &= 6
\end{aligned}$$

9.

$$\begin{aligned}
\frac{x-3}{4} + \frac{x+1}{4} &= 2 && \text{[combine the fractions]}\\
\frac{x-3+x+1}{4} &= 2 && \text{[simplify]}\\
\frac{2x-2}{4} &= 2 && \text{[multiply by 4]}\\
2x-2 &= 8 && \text{[add 2]}\\
2x &= 10 && \text{[divide by 2]}\\
\therefore \quad x &= 5
\end{aligned}$$

10.

$$\begin{aligned}
\frac{2x+5}{3} &= \frac{x-1}{2} && \text{[multiply by 3]}\\
2x+5 &= 3\left(\frac{x-1}{2}\right) && \text{[multiply by 2]}\\
2(2x+5) &= 3(x-1) && \text{[multiply out the brackets]}\\
4x+10 &= 3x-3 && \text{[subtract } 3x\text{]}\\
x+10 &= -3 && \text{[subtract 10]}\\
\therefore \quad x &= -13
\end{aligned}$$

11.

$$\begin{aligned}
\frac{3y+4}{2} - \frac{y-2}{3} &= 5 && \text{[multiply by 2]}\\
3y+4-2\left(\frac{y-2}{3}\right) &= 10 && \text{[multiply by 3]}\\
3(3y+4)-2(y-2) &= 30 && \text{[multiply out the brackets]}\\
9y+12-2y+4 &= 30 && \text{[simplify]}\\
7y+16 &= 30 && \text{[subtract 16]}\\
7y &= 14 && \text{[divide by 7]}\\
\therefore \quad y &= 2
\end{aligned}$$

12.

$$\begin{aligned}
\frac{x+2}{x} &= 3 && \text{[multiply by } x\text{]}\\
x+2 &= 3x && \text{[subtract } 3x\text{]}\\
-2x+2 &= 0 && \text{[subtract 2]}\\
-2x &= -2 && \text{[divide by } -2\text{]}\\
\therefore \quad x &= 1
\end{aligned}$$

13.

$$\begin{aligned}
\frac{2x-1}{x+3} &= 1 && \text{[multiply by } x+3\text{]}\\
2x-1 &= 1(x+3) && \text{[multiply out the brackets]}\\
2x-1 &= x+3 && \text{[subtract } x\text{]}\\
x-1 &= 3 && \text{[add 1]}\\
\therefore \quad x &= 4
\end{aligned}$$

14.

$$
\begin{aligned}
\frac{a+5}{4} + \frac{a-1}{6} &= 2 && \text{[multiply by 4]} \\
a + 5 + 4\left(\frac{a-1}{6}\right) &= 8 && \text{[multiply by 6]} \\
6(a+5) + 4(a-1) &= 48 && \text{[multiply out the brackets]} \\
6a + 30 + 4a - 4 &= 48 && \text{[simplify]} \\
10a + 26 &= 48 && \text{[subtract 26]} \\
10a &= 22 && \text{[divide by 10]} \\
\therefore \quad a &= 2.2
\end{aligned}
$$